香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

Wenchao Miao

Outline Chapter 5 Laplace Transform

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1. Introduction of Laplace Transform

Pierre-Simon, Marquis de Laplace

- 1749–1827 French mathematician and astronomer
- Laplace, along with Lagrange and Legendre of his time, was known as one of the "Three L's" of France
- In 1779, he introduced his transformation method to solve differential equations, but it was not until a century later that this method gained widespread recognition and popularity

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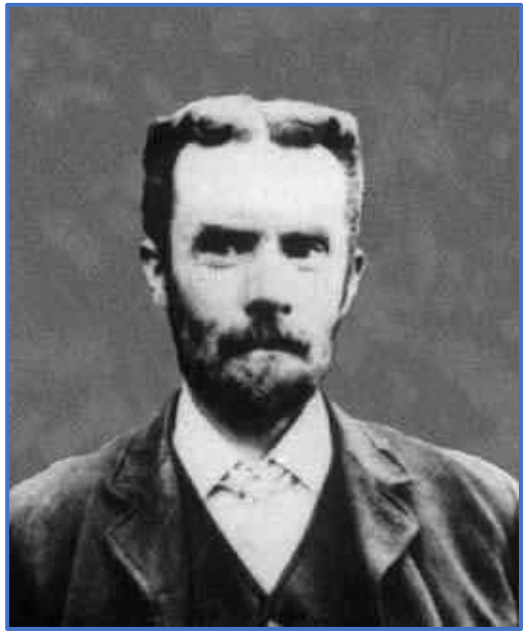
1. Introduction of Laplace Transform

Oliver Heaviside



- 1850–1925 British engineer
- Self-taught with only elementary education
- First to derive the distortionless transmission line solution
- First to write Maxwell's equations in their modern form
- In the late 19th century, developed the "operational calculus," which coincidentally aligned with the Laplace transform

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1. Introduction of Laplace Transform

Advantages:

- **Simplifies problem-solving**, especially for differential equations of systems, **as initial conditions are automatically incorporated**, making it more widely applicable
- Converts differentiation/integration into multiplication/division: Exponential and transcendental functions are transformed into elementary algebraic expressions
- Establishes the concept of **transfer function** based on the convolution theorem, facilitating system analysis

Disadvantages:

- The **physical** interpretation is **less intuitive** compared to the Fourier transform

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1. Introduction of Laplace Transform

Fundamentals	Applications
• Direct Transform • Inverse Transform • Properties	• Solving Differential Equations • Analyzing Dynamic Circuits • System Function: Analyzing system characteristics and block diagrams

Main Content

1. Definition of Laplace Transform
2. Region of Convergence (ROC)

1. Definition of Laplace Transform

Fourier Transform

$$\mathscr{F}[f(t)] = \int_{-\infty}^{\infty} f(t) \cdot \mathrm{e}^{-\mathrm{j}\omega t} \, \mathrm{d}t$$

$f(t)$ may not satisfy the absolute integrability condition

$f(t)$ is multiplied by an exponential decay factor $\mathrm{e}^{-\sigma t}$, σ is a real number

By choosing an appropriate value for σ, $f(t)\mathrm{e}^{-\sigma t}$ can more easily meet the absolute integrability requirement

1. Definition of Laplace Transform

$$\mathscr{F}\left[f(t)\cdot \mathrm{e}^{-\sigma t}\right]=\int_{-\infty}^{\infty}\left[f(t)\mathrm{e}^{-\sigma t}\right]\cdot \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t$$

$$=\int_{-\infty}^{\infty} f(t)\cdot \mathrm{e}^{-(\sigma+\mathrm{j}\omega)t}\,\mathrm{d}t$$

$$=F\left(\sigma+\mathrm{j}\omega\right)$$

Let $\sigma+\mathrm{j}\omega=s$, has the dimension of frequency and is referred to as the complex frequency, then

$$F\left(s\right)=\mathscr{L}\left[f\left(t\right)\right]$$

$$=\int_{-\infty}^{\infty} f\left(t\right)\mathrm{e}^{-st}\,\mathrm{d}t$$

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1. Definition of Laplace Transform

Laplace Transform (Forward Transform)

$$F(s) = \mathscr{L}\left[f(t)\right] = \mathscr{F}\left[f(t)\cdot e^{-\sigma t}\right] = F(\sigma + j\omega)$$

Inverse Fourier transform $F(\sigma + j\omega)$ **yields** $f(t)\cdot e^{-\sigma t}$

$$f(t)e^{-\sigma t} = \mathscr{F}^{-1}\left[F(\sigma + j\omega)\right]$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\sigma + j\omega)e^{j\omega t}\,d\omega$$

Multiplying both sides by $e^{\sigma t}$ **gives the inverse Laplace transform:**

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\sigma + j\omega)e^{(\sigma + j\omega)t}\,d\omega$$

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1. Definition of Laplace Transform

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\sigma+\mathrm{j}\omega)\mathrm{e}^{(\sigma+\mathrm{j}\omega)t}\,\mathrm{d}\omega$$

By substituting $s=\sigma+\mathrm{j}\omega$, we obtain:

$$f(t)=\frac{1}{2\pi\mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty}F(s)\mathrm{e}^{st}\,\mathrm{d}s$$

σ is a constant

Integration limits: for ω is $\int_{-\infty}^{\infty}$, for s is $\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty}$

The path of integration is a vertical line within the region of convergence (ROC)

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1. Definition of Laplace Transform

$$\begin{cases} F(s) = \mathscr{L}\left[f(t)\right] = \int_{-\infty}^{\infty} f(t)\mathrm{e}^{-st}\,\mathrm{d}t \\ f(t) = \mathscr{L}^{-1}\left[F(s)\right] = \dfrac{1}{2\pi\mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} F(s)\mathrm{e}^{st}\,\mathrm{d}s \end{cases}$$

σ?

Bilateral (Two-sided) Laplace Transform

Inverse Laplace Transform

These equations represent the generalized Fourier transform, also known as the bilateral Laplace transform

Where $s = \sigma + \mathrm{j}\omega$, σ a predetermined real constant, ω is the frequency variable

Notation: $f(t) \xleftrightarrow{LT} F(s)$

2. Region of Convergence (ROC)

$s = \sigma + \mathrm{j}\omega$, σ predetermined real constant

σ must be chosen such that the product $f(t) \cdot \mathrm{e}^{-\sigma t}$ converges absolutely

Region of Convergence: the set of all s values for which the Laplace transform $F(s)$ exists

Notation: Abbreviated as ROC (region of convergence)

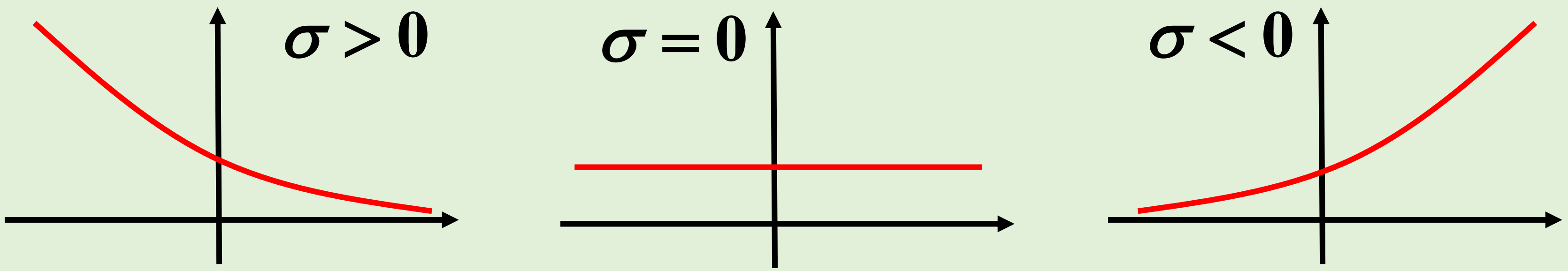
Defines the existence condition for the Laplace transform

2. Definition of Laplace Transform

2. Region of Convergence (ROC)

When analyzing a signal $f(t)$, the Laplace transform applies a exponential damping factor $e^{-\sigma t}$. The behavior of this damping factor depends on the value of σ, and attenuation is unidirectional

Waveform Characteristics of $e^{-\sigma t}$ vs Time t

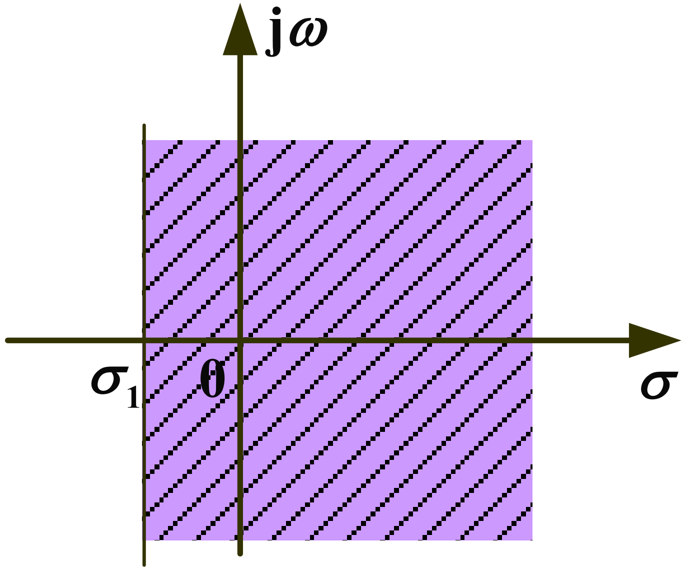


2. Region of Convergence (ROC)

(1) If $f(t)$ is a **right-sided** signal, and for all real numbers $\sigma > \sigma_1$

$$\lim_{t\to\infty} f(t)\cdot e^{-\sigma t} = 0$$

$$\mathrm{ROC:Re}[s] = \sigma > \sigma_1$$



Complex plane $(s\ \text{plane})$: $s = \sigma + j\omega$

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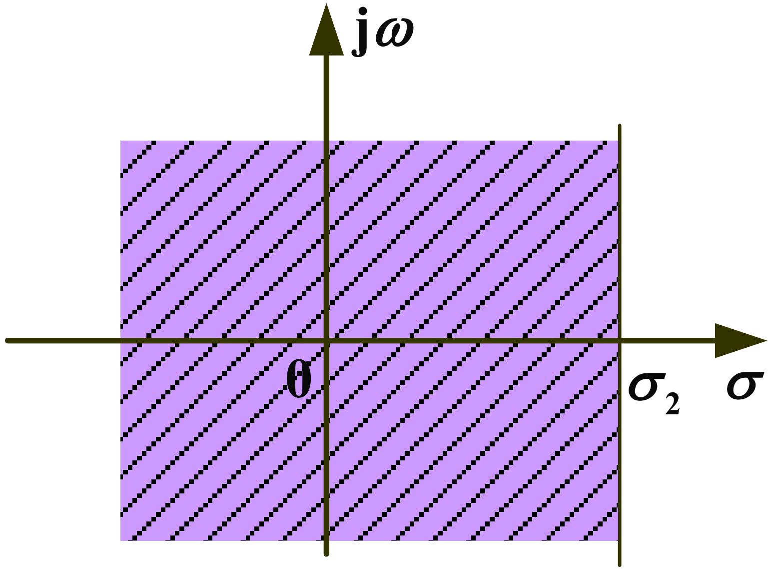
2. Region of Convergence (ROC)

(2) If $f(t)$ is a **left-sided** signal, and for all real numbers $\sigma < \sigma_2$

$$\lim_{t \to -\infty} f(t) \cdot e^{-\sigma t} = 0$$

$$\text{ROC:Re}[s] = \sigma < \sigma_2$$



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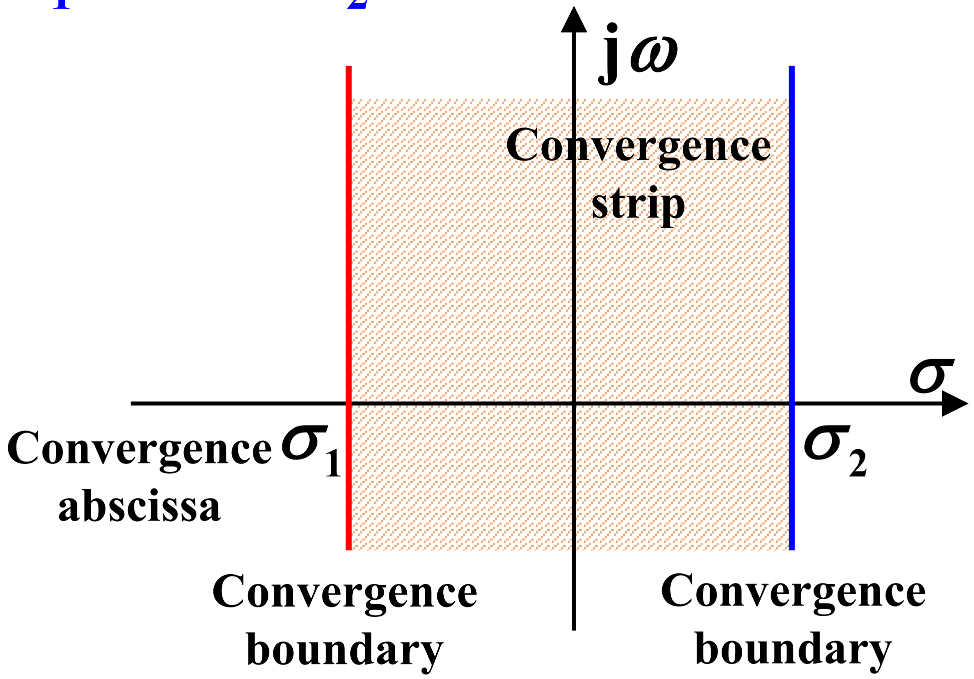
2. Region of Convergence (ROC)

(3) For a two-sided signal, the ROC is the intersection of the ROCs from two one-sided signals

ROC: $\sigma_1 < \sigma < \sigma_2$



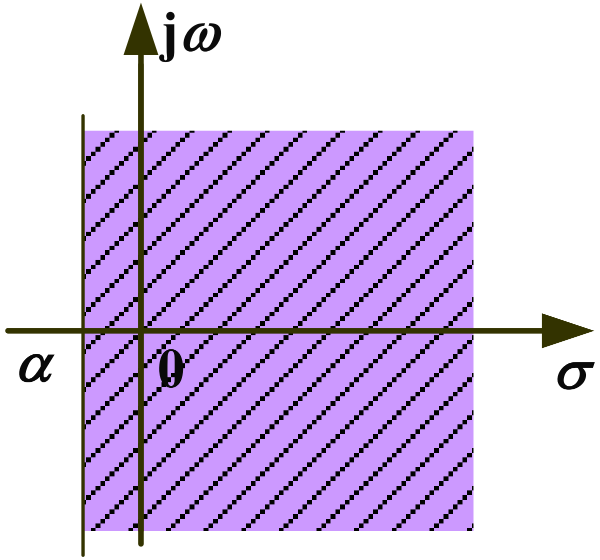
Example 1

Determine the ROC for the signal $f(t)=e^{\alpha t}u(t)$, where $\alpha < 0$

$$\lim_{t\to\infty} e^{\alpha t}\cdot e^{-\sigma t} = \lim_{t\to\infty} e^{(\alpha-\sigma)t} = 0$$

$$\alpha - \sigma < 0$$

ROC: $\sigma > \alpha$



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Example 2

Find the ROC for the Laplace transform of the signal, $f(t) = e^{t^2} u(t)$

$$\lim_{t\to\infty} e^{t^2} \cdot e^{-\sigma t} = \lim_{t\to\infty} e^{\left(t^2 - \sigma t\right)} \neq 0$$

The Laplace transform does not exist (diverges for all σ)

2. Region of Convergence (ROC)

A signal is said to be of exponential order if it satisfies $\begin{cases} \lim\limits_{t\to\infty} f(t)e^{-\sigma t} = 0\left(\sigma > \sigma_1\right) \\ \lim\limits_{t\to-\infty} f(t)e^{-\sigma t} = 0\left(\sigma < \sigma_2\right) \end{cases}$

The Laplace transform always exists for bounded aperiodic signals

For power functions t^n, which grow slower than exponentials e^t, $\lim\limits_{t\to\infty} t^n e^{-\sigma t} = 0$, $\sigma > 0$

For unilateral exponential signals $e^{\alpha t}$, $\lim\limits_{t\to\infty} e^{\alpha t} e^{-\sigma t} = 0$, $\sigma > \alpha$

Signals like e^{t^2}, which grow faster than exponentials, have no ROC (non-exponential order) and thus no Laplace transform

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Summary

Considering that actual signals are **causal signals** and physically realizable systems are **causal**, this course primarily focuses on the **right-sided unilateral Laplace transform**

The region of convergence (ROC) **does not include any poles**, and generally, there **is no need to explicitly note the ROC**

$$\begin{cases} F(s) = \mathscr{L}\left[f(t)\right] = \int_{0-}^{\infty} f(t)\mathrm{e}^{-st}\,\mathrm{d}t \\ f(t) = \mathscr{L}^{-1}\left[F(s)\right] = \dfrac{1}{2\pi \mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} F(s)\mathrm{e}^{st}\,\mathrm{d}s \end{cases}$$

Here, $f(t)$ is referred to as the **original function**, and $F(s)$ is referred to as the **image function**

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Main Content

Laplace Transform of Common Signals

1. Step Function
2. Exponential Function
3. Unit Impulse Signal
4. $t^n u(t)$

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1. Step Function

$$\mathscr{L}\left[u(t)\right] = \int_{0_-}^{\infty} 1 \cdot \mathrm{e}^{-st} \mathrm{d}\, t$$

$$= \frac{1}{-s} \mathrm{e}^{-st} \Big|_{0_-}^{\infty} = \frac{1}{s}$$

Let $\lim\limits_{t\to\infty} \mathrm{e}^{-\sigma t} = 0$

ROC: $\mathrm{Re}[s] > 0$ or $\sigma > 0$

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2. Exponential Function

$$\mathscr{L}\left[\mathrm{e}^{-s_0 t} u(t)\right] = \int_{0_-}^{\infty} \mathrm{e}^{-s_0 t} \mathrm{e}^{-st} \mathrm{d}t \qquad \left(s_0 = \alpha + \mathrm{j}\omega_0\right)$$

$$= \left.\frac{\mathrm{e}^{-(s_0+s)t}}{-(s_0+s)}\right|_{0_-}^{\infty} = \frac{1}{s_0+s}$$

Let $\lim_{t\to\infty} \mathrm{e}^{-s_0 t} \cdot \mathrm{e}^{-\sigma t} = 0$ thus $\lim_{t\to\infty} \mathrm{e}^{-(\alpha+\sigma)t} = 0$

ROC： $\sigma > -\alpha$

3. Unit Impulse Signal

$$\mathscr{L}\left[\delta(t)\right] = \int_{0_-}^{\infty} \delta(t) \cdot \mathrm{e}^{-st} \mathrm{d}t$$

$$= 1$$

ROC: The entire *s*-plane converges

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4. $t^n u(t)$

Integration by parts

$$\mathscr{L}\left[t^n u(t)\right] = \int_{0_-}^{\infty} t^n \cdot e^{-st} \, dt$$

$$= \int_{0_-}^{\infty} t^n \frac{1}{-s} \, de^{-st}$$

$$= \left.\frac{e^{-st}}{-s} t^n \right|_{0_-}^{\infty} - \frac{n}{-s} \int_{0^-}^{\infty} t^{n-1} e^{-st} \, dt \quad = \frac{n}{s} \int_{0_-}^{\infty} t^{n-1} e^{-st} \, dt$$

Let $\lim\limits_{t \to \infty} t^n e^{-\sigma t} = 0$ ROC: $\sigma > 0$

Thus $\mathscr{L}\left[t^n u(t)\right] = \frac{n}{s} \mathscr{L}\left[t^{n-1} u(t)\right]$

4. $t^n u(t)$

$$\mathscr{L}\left[t^n u(t)\right] = \frac{n}{s}\mathscr{L}\left[t^{n-1}u(t)\right]$$

$n = 0 \quad t^n u(t) = u(t)$

$$\mathscr{L}\left[u(t)\right] = \frac{1}{s}$$

$n = 1$

$$\mathscr{L}\left[tu(t)\right] = \frac{1}{s}\left(\frac{1}{s}\right) = \frac{1}{s^2}$$

$n = 2$

$$\mathscr{L}\left[t^2 u(t)\right] = \frac{2}{s}\mathscr{L}\left[tu(t)\right] = \frac{2}{s}\cdot\frac{1}{s^2} = \frac{2}{s^3}$$

$n = 3$

$$\mathscr{L}\left[t^3 u(t)\right] = \frac{3}{s}\mathscr{L}\left[t^2 u(t)\right] = \frac{3}{s}\cdot\frac{2}{s^3} = \frac{6}{s^4}$$

$\cdots\cdots$

Thus $\mathscr{L}\left[t^n u(t)\right] = \dfrac{n!}{s^{n+1}}$ ROC: $\sigma > 0$

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Summary

Laplace Transform of Common Signals

1. Step Function
2. Exponential Function
3. Unit Impulse Signal
4. $t^n u(t)$

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Main Content

1. Linearity Property
2. Shifting Property
3. s-Domain Shift Property
4. Scaling Property
5. Convolution Theorem
6. Differentiation Property
7. Integration Property
8. Initial Value Theorem
9. Final Value Theorem
10. s-Domain Differentiation Property
11. s-Domain Integration Property

Note: Unless otherwise specified, all refer to unilateral transforms

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1. Linearity Property

IF $\mathscr{L}\left[f_1(t)\right] = F_1(s)$

$\mathscr{L}\left[f_2(t)\right] = F_2(s)$

K_1, K_2 constants, thus

$$\mathscr{L}\left[K_1 f_1(t) + K_2 f_2(t)\right] = K_1 F_1(s) + K_2 F_2(s)$$

- The ROC is the **intersection** of the individual ROCs
- The ROC does not include any poles

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Example 1

$$f(t) = \cos(\omega_0 t) u(t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right) u(t)$$

Given

$$\mathscr{L}\left[e^{-s_0 t} u(t)\right] = \frac{1}{s + s_0} \qquad (s_0 \text{ is a complex number})$$

Thus

$$\mathscr{L}\left[\cos(\omega_0 t) u(t)\right] = \frac{1}{2}\left(\frac{1}{s - j\omega_0} + \frac{1}{s + j\omega_0}\right) = \frac{s}{s^2 + \omega_0^2}$$

Similarly

$$\mathscr{L}\left[\sin(\omega_0 t) u(t)\right] = \frac{\omega_0}{s^2 + \omega_0^2}, (\sigma > 0)$$

Example 1

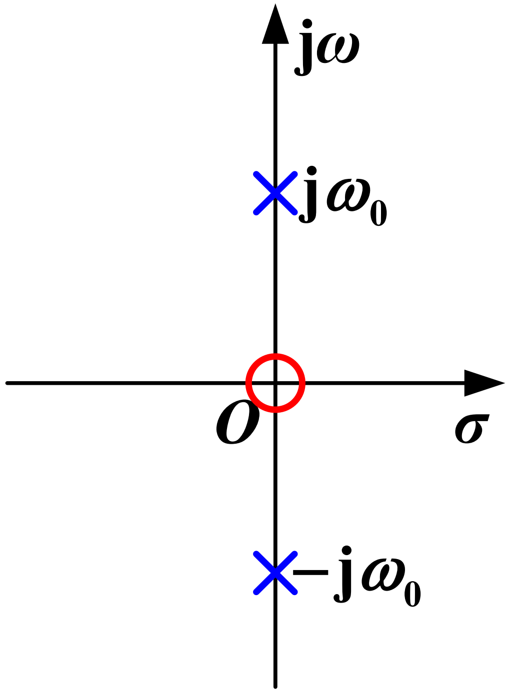
$$\cos(\omega_0 t)u(t) \leftrightarrow F(s) = \frac{s}{s^2 + \omega_0^2}$$

Pole-Zero Plot

- The roots of the denominator polynomial of $F(s)$, denoted as $p_1, p_2, \cdots, p_n$, are called the poles of $F(s)$
- The roots of the numerator polynomial of $F(s)$, denoted as $z_1, z_2, \cdots, z_n$, are called the zeros of $F(s)$

On the s-plane, plot the pole-zero diagram of $F(s)$:

- Poles: Represented by × (cross marks)
- Zeros: Represented by ○ (open circles)



2. Shifting Property

If $L[f(t)] = F(s)$,

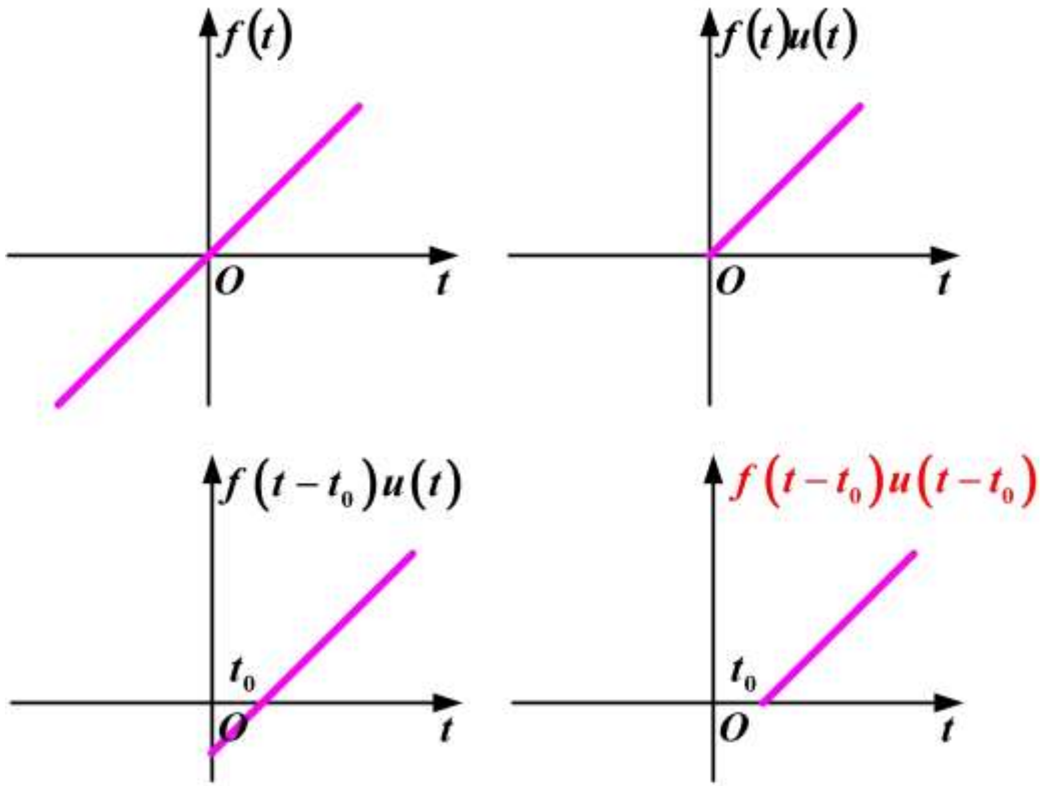
Then the Laplace transform of $f(t-t_0)u(t-t_0)$ **(where** $t_0 > 0$**) is:**

$$F(s)e^{-st_0}$$

$$f(t) \overset{Unilateral\,Transform}{\longrightarrow} f(t)u(t) \overset{Time\,Shift}{\longrightarrow} f(t-t_0)u(t-t_0)$$

$$\begin{cases} t \to t - t_0 \\ \text{determine time shift based on } u(t) \end{cases}$$

2. Shifting Property



$$\begin{cases} t \to t - t_0 \\ \text{determine time shift based on } u(t) \end{cases}$$

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2. Shifting Property

Proof $\mathscr{L}\left[f(t-t_0)u(t-t_0)\right]=\int_{0_-}^{\infty} f(t-t_0)u(t-t_0)\mathrm{e}^{-st}\mathrm{d}t$

Let $\tau=t-t_0$ $\quad =\int_{t_0}^{\infty} f(t-t_0)\mathrm{e}^{-st}\mathrm{d}t$

Then $t=\tau+t_0$ $\quad =\mathrm{e}^{-st_0}\int_{0_-}^{\infty} f(\tau)\mathrm{e}^{-s\tau}\mathrm{d}\tau$

$\mathrm{d}t=\mathrm{d}\tau$, **Substitute into the above equation** $\quad =F(s)\mathrm{e}^{-st_0}$

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Example 2

Find the Laplace transform of $f(t) = e^{-t}u(t-2)$

$$f(t) = e^{-2}e^{-(t-2)}u(t-2)$$

Given $e^{-t}u(t) \leftrightarrow \frac{1}{s+1}$

$$F(s) = e^{-2} \cdot \frac{1}{s+1} \cdot e^{-2s}$$

$$= \frac{1}{s+1} \cdot e^{-2(s+1)}$$

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3. s-Domain Shift Property

If $\mathscr{L}\left[f(t)\right]=F(s)$,

Then $\mathscr{L}\left[f(t)\mathrm{e}^{-\alpha t}\right]=F(s+\alpha)$

Proof

$$\mathscr{L}\left[f(t)\mathrm{e}^{-\alpha t}\right]=\int_{0_-}^{\infty}f(t)\mathrm{e}^{-\alpha t}\mathrm{e}^{-st}\,\mathrm{d}t$$

$$=F(s+\alpha)$$

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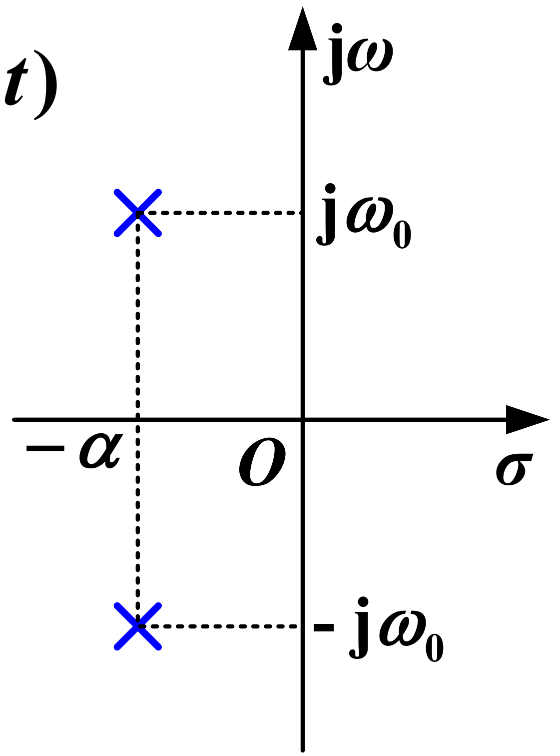
Example 3

Find the Laplace transform of $e^{-\alpha t}\cos(\omega_0 t)u(t)$

Given: $\mathscr{L}\left[\cos\left(\omega_0 t\right)u(t)\right] = \dfrac{s}{s^2+\omega_0^2}$

Thus $e^{-\alpha t}\cos\left(\omega_0 t\right)u(t) \xleftrightarrow{LT} \dfrac{s+\alpha}{\left(s+\alpha\right)^2+\omega_0^2}$

Similarly : $e^{-\alpha t}\sin\left(\omega_0 t\right)u(t) \xleftrightarrow{LT} \dfrac{\omega_0}{\left(s+\alpha\right)^2+\omega_0^2}$



ω_0: *angular frequency*

$-\alpha$: *amplitude*

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4. Scaling Property

$$\text{If } \mathscr{L}\left[f(t)\right]=F(s),\quad \text{Then } \mathscr{L}\left[f(at)\right]=\frac{1}{a}F\left(\frac{s}{a}\right)\qquad (a>0)$$

When both time shifting and time scaling are involved:

$$\mathscr{L}\left[f(at-b)u(at-b)\right]=\frac{1}{a}F\left(\frac{s}{a}\right)\mathrm{e}^{-s\frac{b}{a}}\qquad (a>0, b>0)$$

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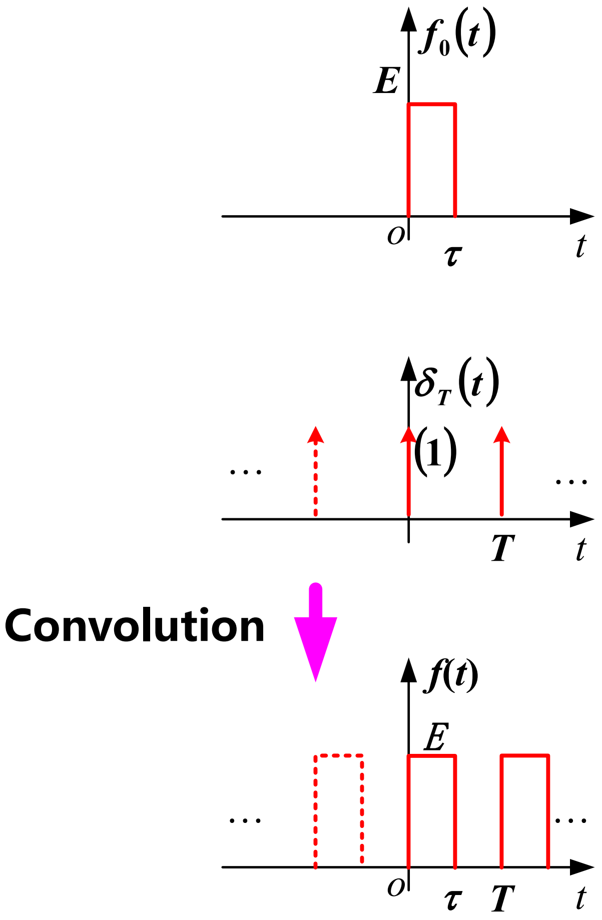
5. Convolution Theorem

If $\mathscr{L}\left[f_1(t)\right]=F_1(s)$, $\mathscr{L}\left[f_2(t)\right]=F_2(s)$, $f_1(t), f_2(t)$ are causal signals, then:

$$\mathscr{L}\left[f_1(t) * f_2(t)\right]=F_1(s)F_2(s)$$

$$\mathscr{L}\left[f_1(t)\cdot f_2(t)\right]=\frac{1}{2\pi \mathrm{j}}F_1(s) * F_2(s)$$

Example 4



$$\mathscr{L}\left[\delta_T(t)\right] = 1 + e^{-sT} + e^{-s2T} + \cdots$$

$$= \frac{1}{1 - e^{-sT}}$$

For a general signal $f(t) = f_0(t) * \delta_T(t)$

$$F(s) = F_0(s)\frac{1}{1 - e^{-sT}}$$

$\frac{1}{1 - e^{-sT}}$ Termed the "Periodization Factor"

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6. Differentiation Property

(1) Differentiation Property for Bilateral Laplace Transform

$$\text{If } f(t) \xleftrightarrow{\mathscr{L}} F(s), \text{ then } \frac{\mathrm{d}f(t)}{\mathrm{d}t} \xleftrightarrow{\mathscr{L}} sF(s)$$

Proof:

The inverse bilateral Laplace transform is given by:

$$f(t) = \frac{1}{2\pi\mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} F(s)\mathrm{e}^{st}\,\mathrm{d}s$$

Differentiating both sides with respect to t:

$$\frac{\mathrm{d}f(t)}{\mathrm{d}t} = \frac{1}{2\pi\mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} F(s)\,s\,\mathrm{e}^{st}\,\mathrm{d}s$$

Application: System function analysis

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6. Differentiation Property

(2) Differentiation Property for Unilateral Laplace Transform (for reference)

$$\text{If } f(t) \overset{\mathscr{L}}{\longleftrightarrow} F(s), \quad \text{then } \frac{\mathrm{d}\, f(t)}{\mathrm{d}\, t} \overset{\mathscr{L}}{\longleftrightarrow} sF(s) - f(0_-)$$

Proof:

$$\int_{0_-}^{\infty} f'(t)\mathrm{e}^{-st}\,\mathrm{d}t = \int_{0_-}^{\infty} \mathrm{e}^{-st}\,\mathrm{d}f(t)$$

$$= f(t)\mathrm{e}^{-st}\Big|_{0_-}^{\infty} - \int_{0_-}^{\infty} f(t)\,\mathrm{d}\mathrm{e}^{-st}$$

$$= -f(0_-) - \int_{0_-}^{\infty} (-s) f(t)\mathrm{e}^{-st}\,\mathrm{d}t$$

$$= -f(0_-) + sF(s)$$

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6. Differentiation Property

$$\frac{\mathrm{d}\,f(t)}{\mathrm{d}\,t} \xleftrightarrow{\mathscr{L}} sF(s) - f(0_-)$$

$$\frac{\mathrm{d}\,f^2(t)}{\mathrm{d}\,t} \xleftrightarrow{\mathscr{L}} s\left[sF(s) - f(0_-)\right] - f'(0_-)$$

$$= s^2F(s) - sf(0_-) - f'(0_-)$$

$$\frac{\mathrm{d}\,f^n(t)}{\mathrm{d}\,t} \xleftrightarrow{\mathscr{L}} s^nF(s) - \sum_{r=0}^{n-1} s^{n-r-1} f^{(r)}(0_-)$$

Application:

(1) Solving differential equation

(2) Dynamic circuit analysis

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7. Integration Property

(1) Integration Property for Bilateral Laplace Transform

If $f(t) \xleftrightarrow{\mathscr{L}} F(s)$, then $\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau \xleftrightarrow{\mathscr{L}} \dfrac{F(s)}{s}$

Proof:

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau = f(t) * u(t)$$

$$u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s}$$

Using the time-domain convolution theorem, obtain

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau \xleftrightarrow{\mathscr{L}} \frac{F(s)}{s}$$

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7. Integration Property

(2) Integration Property for Unilateral Laplace Transform (for reference)

If $f(t) \xleftrightarrow{\mathscr{L}} F(s)$, then

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau \xleftrightarrow{\mathscr{L}} \frac{F(s)}{s} + \frac{f^{(-1)}(0_-)}{s}$$

Proof:

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau = \underbrace{\int_{-\infty}^{0_-} f(\tau)\,\mathrm{d}\tau}_{①} + \underbrace{\int_{0_-}^{t} f(\tau)\,\mathrm{d}\tau}_{②}$$

① $f^{(-1)}(t) = \int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau$ ie $\int_{-\infty}^{0_-} f(\tau)\,\mathrm{d}\tau = f^{(-1)}(0_-)$

So, Laplace transform of ① is $\mathscr{L}\left[f^{(-1)}(0_-)\right] = \dfrac{f^{(-1)}(0_-)}{s}$

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7. Integration Property

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau = \underbrace{\int_{-\infty}^{0_-} f(\tau)\,\mathrm{d}\tau}_{①} + \underbrace{\int_{0_-}^{t} f(\tau)\,\mathrm{d}\tau}_{②}$$

The LT of ② is $\int_{0_-}^{\infty}\left[\int_{0_-}^{t} f(\tau)\,\mathrm{d}\tau\right]\mathrm{e}^{-st}\,\mathrm{d}t$

$$= \int_{0_-}^{\infty}\left[\int_{0_-}^{\infty} f(\tau)u(t-\tau)\,\mathrm{d}\tau\right]\mathrm{e}^{-st}\,\mathrm{d}t$$

$$= \int_{0_-}^{\infty} f(\tau)\int_{0_-}^{\infty} u(t-\tau)\mathrm{e}^{-st}\,\mathrm{d}t\,\mathrm{d}\tau$$

$$= \frac{1}{s}\int_{0_-}^{\infty} f(\tau)\mathrm{e}^{-s\tau}\,\mathrm{d}\tau = \frac{F(s)}{s}$$

8. Initial Value Theorem

If $f(t)$ and $\frac{df(t)}{dt}$ are Laplace-transformable, and $f(t) \leftrightarrow F(s)$, then:

$$lim_{t\to 0_+} f(t) = f(0_+) = lim_{s\to\infty} sF(s)$$

Notes:

1. If $F(s)$ is not a proper fraction, express it as a polynomial plus a proper fraction:

$$F(s) = F_1(s) + \cdots + k_1 s + k_0$$

2. The initial value is determined by:

$$f(0_+) = lim_{s\to\infty} sF_1(s)$$

3. For the polynomial part:

$$f_2(t) = \cdots k_1\, s + k_0 \leftrightarrow \cdots + k_1\delta'(t) + k_0\delta(t)$$

$$f_2(0_+) = 0$$

But $lim_{s\to\infty} s(\cdots + k_1 s + k_0) = \infty$

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8. Initial Value Theorem

Proof:

From the time-domain differentiation property:

Using the 0_- system: $$\int_{0_-}^{\infty} \frac{\mathrm{d}f(t)}{\mathrm{d}t} e^{-st} \mathrm{d}t = sF(s) - f(0_-)$$

Using the 0_+ system: $$\int_{0_+}^{\infty} \frac{\mathrm{d}f(t)}{\mathrm{d}t} e^{-st} \mathrm{d}t = sF(s) - f(0_+)$$

$$\lim_{s\to\infty}\left[\int_{0_+}^{\infty} \frac{\mathrm{d}f(t)}{\mathrm{d}t} e^{-st} \mathrm{d}t\right] = \int_{0_+}^{\infty} \frac{\mathrm{d}f(t)}{\mathrm{d}t}\left[\lim_{s\to\infty} e^{-st}\right]\mathrm{d}t = 0$$

$$\therefore f(0_+) = \lim_{s\to\infty} sF(s)$$

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Example 5

$$F(s)=\frac{2s}{s+1}, \quad f(0_+)=?$$

Because $F(s)=\frac{2s}{s+1}=-\frac{2}{s+1}+2$

The function $f(t)$ contains a term $2\delta(t)$

Let $F_0(s)=-\frac{2}{s+1}$

Thus $f(0_+)=\lim_{s\to\infty}\left[sF_0(s)\right]$

$$=\lim_{s\to\infty} s\frac{-2}{s+1}=\lim_{s\to\infty}\frac{-2}{1+\frac{1}{s}}=-2$$

9. Final Value Theorem

Let $f(t)$ and $\frac{d}{dt}f(t)$ have Laplace transforms, and suppose:

$$L[f(t)] = F(s)$$

Then:

$$f(\infty) = lim_{t\to\infty} f(t) = lim_{s\to 0} sF(s)$$

Conditions for Finite Final Value:

1. All poles in the **left** half-plane (LHP) of s : Final value is 0 (system stabilizes to zero)
2. Single pole at $s = 0$: The final value equals the coefficient k of the $\frac{k}{s}$ term in the **partial fraction expansion**

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9. Final Value Theorem

Proof:

Based on the formula derived from the proof of the Initial Value Theorem

$$sF(s) = f(0_+) + \int_{0_+}^{\infty} \frac{\mathrm{d}\, f(t)}{\mathrm{d}\, t} \mathrm{e}^{-st}\, \mathrm{d}\, t$$

$$\lim_{s\to 0} sF(s) = f(0_+) + \lim_{s\to 0} \int_{0_+}^{\infty} \frac{\mathrm{d}\, f(t)}{\mathrm{d}\, t} \mathrm{e}^{-st}\, \mathrm{d}\, t$$

$$= f(0_+) + \lim_{t\to\infty} f(t) - f(0_+)$$

$$= \lim_{t\to\infty} f(t)$$

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9. Final Value Theorem

Conditions for the Existence of a Final Value

When poles are in the left half-plane (LHP) of s, the final value is 0

Single Pole $\quad \dfrac{1}{s+2} \leftrightarrow e^{-2t}u(t)$

Complex Conjugate Poles $\quad \dfrac{s+2}{(s+2)^2+\omega_1^2} \leftrightarrow e^{-2t}\cos(\omega_1 t)u(t)$

Double Pole $\quad \dfrac{1}{(s+2)^2} \leftrightarrow te^{-2t}u(t)$

When poles are in the right half-plane (RHP) of s, the original function diverges $\quad \dfrac{1}{s-2} \leftrightarrow e^{+2t}u(t)$

9. Final Value Theorem

Conditions for the Existence of a Final Value

When poles are on the axis of the s-plane

Single pole at origin	$\frac{1}{s} \leftrightarrow u(t)$	Final value is 1
Multiple poles at origin	$\frac{1}{s^2} \leftrightarrow tu(t)$	No finite final value
Complex conjugate poles on imaginary axis	$\frac{s}{s^2+\omega_1^2} \leftrightarrow \cos(\omega_1 t)u(t)$	Oscillates indefinitely (no final value)

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9. Final Value Theorem

There are only two cases where a finite final value exists

(1) All poles in the left half-plane (LHP) of s, Final value = 0

(2) Single pole at $s = 0$, The final value equals the coefficient k of the k/s term in the partial fraction expansion, calculated as:

$ku(t)$

$$k = sF(s)\big|_{s=0}$$

$$f(\infty) = \lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

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10. s-Domain Differentiation Property

$$\text{If } \mathscr{L}\left[f(t)\right]=F(s), \quad \text{then } \mathscr{L}\left[tf(t)\right]=-\frac{\mathrm{d}F(s)}{\mathrm{d}s}$$

Generalization: $$\mathscr{L}\left[t^{n}f(t)\right]=(-1)^{n}\frac{\mathrm{d}^{n}F(s)}{\mathrm{d}s^{n}}$$

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Summary

1. Linearity Property
2. Shifting Property
3. s-Domain Shift Property
4. Scaling Property
5. Convolution Theorem
6. Differentiation Property
7. Integration Property
8. Initial Value Theorem
9. Final Value Theorem
10. s-Domain Differentiation Property

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Summary

Signal	Time domain	S-domain
Step function	$u(t)$	$\frac{1}{s}$
exponential function	$e^{-s_0 t}u(t)$	$\frac{1}{s+s_0}$
Unit impulse signal	$\delta(t)$	1
$tu(t)$	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
cos	$\cos(w_0 t)\ u(t)$	$\frac{s}{s^2+w_0^2}$
sin	$\sin(w_0 t)\ u(t)$	$\frac{w_0}{s^2+w_0^2}$

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Summary

Property	$f(t)$	$F(s)$
Linearity	$K_1 f_1(t) + K_2 f_2(t)$	$K_1 F_1(t) + K_2 F_2(t)$
Time shifting	$f(t - t_0)u(t - t_0)$	$F(s)e^{-st_0}$
s-domain shift	$f(t)e^{-\alpha t}$	$F(s + \alpha)$
Scaling	$f(at)$	$\frac{1}{a}F(\frac{s}{a})$
Time shifting + scaling	$f(at - b)u(at - b)$	$\frac{1}{a}F(\frac{s}{a})e^{-s\frac{b}{a}}$
Convolution	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Bilateral differentiation	$\frac{df(t)}{dt}$	$sF(s)$

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Summary

Unilateral differentiation	$\frac{df(t)}{dt}$	$sF(s) - f(0_-)$
Bilateral integral	$\int_{-\infty}^{t} f(\tau)d\tau$	$\frac{F(s)}{s}$
Unilateral integral	$\int_{-\infty}^{t} f(\tau)d\tau$	$\frac{F(s)}{s} + \frac{f^{-1}(0_-)}{s}$
Initial value	$f(0_+)$	$\lim_{s\to\infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s\to 0} sF(s)$
s-domain differentiation	$t^n f(t)$	$(-1)^n \frac{d^n F(s)}{ds^n}$
s-domain integration	$\frac{f(t)}{t}$	$\int_{s}^{\infty} F(\lambda)d\lambda$